

ECON651 Advanced Behavioral Economics

Fall 2026

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Description

This is a second course in Behavioral Economics, which is independent of but complementary to ECON650. Some topics to be explored are: the characterization of rational behavior and its empirical testing; the framework of bounded rationality; models of limited attention, consideration sets and rational inattention; reference points and other contextual effects; “the cognitive turn in behavioral economic” and “cognitive economics”, cognitive noise and cognitive uncertainty; similar developments in the cognitive sciences, resource-rational analysis; simplicity and complexity. Students will also be presented with recent applications of these topics to areas such as macroeconomics and economic development.

Evaluation

1. 50% Presentation
2. 50% Research Proposal

Students will be evaluated, with equal weights, based on an in-class presentation and a research proposal.

The topics (and date) for the presentation are to be discussed with the instructor in advance. They can range from one particular paper to a broader literature, and from theoretical to applied papers. Students will be evaluated on their presentation skills, critical thinking and mastery of the material they present, as well as their capacity to establish connections between their chosen topics and the ideas, theories and methods from the course. Students will have to send their presentation slides to the instructor in advance of their presentation.

Students will also need to submit, by 12pm of December 15th, a research proposal relevant to the issues discussed during the course. Students are encouraged to discuss their ideas with the instructor in advance. The proposal may or may not be connected to the topics from their presentation. The proposal should clearly establish a research question, include a literature review establishing that the question remains unanswered, and discuss the methodology to be used to answer the question. The proposal might also, though it doesn't need to, include initial results towards answering the question.

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Course Policies

Meetings Tuesday and Thursday, 01:30-3:00PM, LT 907

Office Hours Thursday, 03:30PM-04:30PM (additional ones by appointment), LT 1004

Generative AI Policy Students must not use generative AI for developing and writing their research proposal for the course.

Course Topics and Readings

1 Rationality and the Behavioral Benchmark

1.1 Rational Choice and Revealed Preference

Classic

- Samuelson, Paul A. (1938). “A Note on the Pure Theory of Consumer’s Behaviour.” *Economica* 5(17): 61–71.
- Houthakker, H. S. (1950). “Revealed Preference and the Utility Function.” *Economica* 17(66): 159–174.
- Afriat, Sydney N. (1967). “The Construction of Utility Functions from Expenditure Data.” *International Economic Review* 8(1): 67–77.

Mandatory

- Tipoe, Eileen, Abi Adams, and Ian Crawford (2022). “Revealed Preference Analysis and Bounded Rationality.” *Oxford Economic Papers* 74(2): 313–332.
- de Clippel, Geoffroy, and Kareen Rozen (2024). “Bounded Rationality in Choice Theory: A Survey.” *Journal of Economic Literature* 62(3): 995–1039.

Additional

- Dean, Mark, and Daniel Martin (2016). “Measuring Rationality with the Minimum Cost of Revealed Preference Violations.” *Review of Economics and Statistics* 98(3): 524–534.
- Beatty, Timothy K. M., and Ian A. Crawford (2011). “How Demanding Is the Revealed Preference Approach to Demand?” *American Economic Review* 101(6): 2782–2795.
- Choi, Syngjoo, Shachar Kariv, Wieland Müller, and Dan Silverman (2014). “Who Is (More) Rational?” *American Economic Review* 104(6): 1518–1550.
- Nishimura, Hiroki, Efe A. Ok, and John K.-H. Quah (2017). “A Comprehensive Approach to Revealed Preference Theory.” *American Economic Review* 107(4): 1239–1263.

1.2 Stochastic Choice

Classic

- Luce, R. Duncan (1959). *Individual Choice Behavior: A Theoretical Analysis*. New York: Wiley.
- McFadden, Daniel (1974). “Conditional Logit Analysis of Qualitative Choice Behavior.” In Paul Zarembka, ed., *Frontiers in Econometrics*, 105–142. New York: Academic Press.

Mandatory

- Agranov, Marina, and Pietro Ortoleva (2017). “Stochastic Choice and Preferences for Randomization.” *Journal of Political Economy* 125(1): 40–68.
- Natenzon, Paulo (2019). “Random Choice and Learning.” *Journal of Political Economy* 127(1): 419–457.

Additional

- Strzalecki, Tomasz (2025). *Stochastic Choice Theory*. Econometric Society Monographs. Cambridge: Cambridge University Press.
- Kitamura, Yuichi, and Jörg Stoye (2018). “Nonparametric Analysis of Random Utility Models.” *Econometrica* 86(6): 1883–1909.
- McKelvey, Richard D., and Thomas R. Palfrey (1995). “Quantal Response Equilibria for Normal Form Games.” *Games and Economic Behavior* 10(1): 6–38.
- Cerreia-Vioglio, Simone, David Dillenberger, Pietro Ortoleva, and Giuseppe Riella (2019). “Deliberately Stochastic.” *American Economic Review* 109(7): 2425–2445.

2 Classical Bounded Rationality

2.1 Procedural Rationality, Satisficing, and Search

Classic

- Simon, Herbert A. (1955). “A Behavioral Model of Rational Choice.” *Quarterly Journal of Economics* 69(1): 99–118.
- Simon, Herbert A. (1956). “Rational Choice and the Structure of the Environment.” *Psychological Review* 63(2): 129–138.
- Stigler, George J. (1961). “The Economics of Information.” *Journal of Political Economy* 69(3): 213–225.
- Weitzman, Martin L. (1979). “Optimal Search for the Best Alternative.” *Econometrica* 47(3): 641–654.

Mandatory

- Caplin, Andrew, and Mark Dean (2011). “Search, Choice, and Revealed Preference.” *Theoretical Economics* 6(1): 19–48.
- Caplin, Andrew, Mark Dean, and Daniel Martin (2011). “Search and Satisficing.” *American Economic Review* 101(7): 2899–2922.

Additional

- Rubinstein, Ariel (1998). *Modeling Bounded Rationality*. Cambridge, MA: MIT Press.

2.2 Consideration Sets and Limited Attention

Classic

- Masatlioglu, Yusufcan, Daisuke Nakajima, and Erkut Y. Ozbay (2012). “Revealed Attention.” *American Economic Review* 102(5): 2183–2205.
- Manzini, Paola, and Marco Mariotti (2014). “Stochastic Choice and Consideration Sets.” *Econometrica* 82(3): 1153–1176.

Mandatory

- Cattaneo, Matias D., Xinwei Ma, Yusufcan Masatlioglu, and Elchin Suleymanov (2020). “A Random Attention Model.” *Journal of Political Economy* 128(7): 2796–2836.
- Abaluck, Jason, and Abi Adams-Prassl (2021). “What Do Consumers Consider Before They Choose? Identification from Asymmetric Demand Responses.” *Quarterly Journal of Economics* 136(3): 1611–1663.

Additional

- Honka, Elisabeth, Ali Hortaçsu, and Matthijs Wildenbeest (2019). “Empirical Search and Consideration Sets.” In *Handbook of the Economics of Marketing*, Vol. 1, 193–257. Amsterdam: Elsevier.
- Aguiar, Victor H., Maria Jose Boccardi, Nail Kashaev, and Jeongbin Kim (2023). “Random Utility and Limited Consideration.” *Quantitative Economics* 14(1): 71–116.

Choice Overload and Large Menus

Classic

- Iyengar, Sheena S., and Mark R. Lepper (2000). “When Choice Is Demotivating: Can One Desire Too Much of a Good Thing?” *Journal of Personality and Social Psychology* 79(6): 995–1006.

Additional

- Scheibehenne, Benjamin, Rainer Greifeneder, and Peter M. Todd (2010). “Can There Ever Be Too Many Options? A Meta-Analytic Review of Choice Overload.” *Journal of Consumer Research* 37(3): 409–425.
- Chernev, Alexander, Ulf Böckenholt, and Joseph Goodman (2015). “Choice Overload: A Conceptual Review and Meta-Analysis.” *Journal of Consumer Psychology* 25(2): 333–358.
- Bhargava, Saurabh, George Loewenstein, and Justin Sydnor (2017). “Choose to Lose: Health Plan Choices from a Menu with Dominated Option.” *Quarterly Journal of Economics* 132(3): 1319–1372.
- de Lara, Lucas, and Mark Dean “Rational Choice Overload.” Working paper.

3 Cognitive Economics

3.1 From Behavioral Anomalies to Cognitive Mechanisms

Mandatory

- Woodford, Michael (2020). “Modeling Imprecision in Perception, Valuation and Choice.” *Annual Review of Economics* 12: 579–601.
- Lieder, Falk, and Thomas L. Griffiths (2020). “Resource-Rational Analysis: Understanding Human Cognition as the Optimal Use of Limited Computational Resources.” *Behavioral and Brain Sciences* 43: e1.
- Enke, Benjamin (2024). “The Cognitive Turn in Behavioral Economics.” Working paper.

3.2 Noisy Cognition and Imprecise Representation

Mandatory

- Woodford, Michael (2020). “Modeling Imprecision in Perception, Valuation and Choice.” *Annual Review of Economics* 12: 579–601.
- Khaw, Mel Win, Ziang Li, and Michael Woodford (2021). “Cognitive Imprecision and Small-Stakes Risk Aversion.” *Review of Economic Studies* 88(4): 1979–2013.

Additional

- Schley, Dan R., and Ellen Peters (2014). “Assessing ‘Economic Value’: Symbolic-Number Mappings Predict Risky and Riskless Valuations.” *Psychological Science* 25(3): 753–761.
- Petzschner, Frederike H., Stefan Glasauer, and Klaas E. Stephan (2015). “A Bayesian Perspective on Magnitude Estimation.” *Trends in Cognitive Sciences* 19(5): 285–293.
- Zhang, Hang, Xueyi Ren, and Laurence T. Maloney (2020). “The Bounded Rationality of Probability Distortion: The Bounded Log-Odds Model.” *Proceedings of the National Academy of Sciences* 117: 22024–22034.

3.3 Approximate Inference and Information Sampling

Classic

- Knill, David C., and Alexandre Pouget (2004). “The Bayesian Brain: The Role of Uncertainty in Neural Coding and Computation.” *Trends in Neurosciences* 27(12): 712–719.
- Stewart, Neil, Nick Chater, and Gordon D. A. Brown (2006). “Decision by Sampling.” *Cognitive Psychology* 53(1): 1–26.
- Doya, Kenji, Shin Ishii, Alexandre Pouget, and Rajesh P. N. Rao, eds. (2007). *Bayesian Brain: Probabilistic Approaches to Neural Coding*. Cambridge, MA: MIT Press.

Mandatory

- Benjamin, Daniel J. (2019). “Errors in Probabilistic Reasoning and Judgment Biases.” In *Handbook of Behavioral Economics*, Vol. 2, 69–186. Amsterdam: Elsevier.
- Zhu, Jian-Qiao, Adam N. Sanborn, and Nick Chater (2020). “The Bayesian Sampler: Generic Bayesian Inference Causes Incoherence in Human Probability Judgments.” *Psychological Review* 127(5): 719–748.

Additional

- Marchiori, Davide, Sibilla Di Guida, and Ido Erev (2015). “Noisy Retrieval Models of Over- and Under-Sensitivity to Rare Events.” *Decision* 2(2): 82–106.
- Lieder, Falk, Thomas L. Griffiths, and Ming Hsu (2018). “Overrepresentation of Extreme Events in Decision Making Reflects Rational Use of Cognitive Resources.” *Psychological Review* 125(1): 1–32.
- Dasgupta, Ishita, Eric Schulz, Joshua B. Tenenbaum, and Samuel J. Gershman (2020). “A Theory of Learning to Infer.” *Psychological Review* 127(3): 412–441.
- Esponda, Ignacio, Emanuel Vespa, and Sevgi Yuksel (2024). “Mental Models and Learning: The Case of Base-Rate Neglect.” *American Economic Review* 114(3): 752–782.

3.4 Cognitive Uncertainty

Mandatory

- Enke, Benjamin, and Thomas Graeber (2023). “Cognitive Uncertainty.” *Quarterly Journal of Economics* 138(4): 2021–2067.
- Enke, Benjamin, Thomas Graeber, Ryan Oprea, and Jeffrey Yang (2024). “Behavioral Attenuation.” Working paper.

4 Attention and Endogenous Information Acquisition

4.1 Rational Inattention

Classic

- Sims, Christopher A. (2003). “Implications of Rational Inattention.” *Journal of Monetary Economics* 50(3): 665–690.
- Sims, Christopher A. (2006). “Rational Inattention: Beyond the Linear-Quadratic Case.” *American Economic Review* 96(2): 158–163.

Mandatory

- Matějka, Filip, and Alisdair McKay (2015). “Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model.” *American Economic Review* 105(1): 272–298.
- Maćkowiak, Bartosz, Filip Matějka, and Mirko Wiederholt (2023). “Rational Inattention: A Review.” *Journal of Economic Literature* 61(1): 226–273.

Additional

- Fosgerau, Mogens, Emerson Melo, André de Palma, and Matthew Shum (2020). “Discrete Choice and Rational Inattention: A General Equivalence Result.” *International Economic Review* 61(4): 1569–1589.
- Cover, Thomas M., and Joy A. Thomas (2006). *Elements of Information Theory*, 2nd ed. Hoboken, NJ: Wiley. Selected chapters.

4.2 Revealed Rational Inattention

Mandatory

- Caplin, Andrew, and Mark Dean (2015). “Revealed Preference, Rational Inattention, and Costly Information Acquisition.” *American Economic Review* 105(7): 2183–2203.
- de Oliveira, Henrique, Tommaso Denti, Maximilian Mihm, and Kemal Ozbek (2017). “Rationally Inattentive Preferences and Hidden Information Costs.” *Theoretical Economics* 12(2): 621–654.

Additional

- Caplin, Andrew, Dániel Csaba, John Leahy, and Oded Nov (2020). “Rational Inattention, Competitive Supply, and Psychometrics.” *Quarterly Journal of Economics* 135(3): 1681–1724.

4.3 General Information Costs

Mandatory

- Caplin, Andrew, Mark Dean, and John Leahy (2022). “Rationally Inattentive Behavior: Characterizing and Generalizing Shannon Entropy.” *Journal of Political Economy* 130(6): 1676–1715.
- Denti, Tommaso (2022). “Posterior Separable Cost of Information.” *American Economic Review* 112(10): 3215–3259.

Additional

- Hébert, Benjamin, and Michael Woodford (2021). “Neighborhood-Based Information Costs.” *American Economic Review* 111(10): 3225–3255.
- Denti, Tommaso, Massimo Marinacci, and Aldo Rustichini (2022). “Experimental Cost of Information.” *American Economic Review* 112(9): 3106–3123.
- de Oliveira, Henrique, Tommaso Denti, Maximilian Mihm, and Kemal Ozbek (2017). “Rationally Inattentive Preferences and Hidden Information Costs.” *Theoretical Economics* 12(2): 621–654.

4.4 Behavioral Inattention and Sparse Processing

Classic

- Gabaix, Xavier (2014). “A Sparsity-Based Model of Bounded Rationality.” *Quarterly Journal of Economics* 129(4): 1661–1710.

Mandatory

- Gabaix, Xavier (2019). “Behavioral Inattention.” In *Handbook of Behavioral Economics*, Vol. 2, 261–343. Amsterdam: Elsevier.
- Loewenstein, George, and Zachary Wojtowicz (2025). “The Economics of Attention.” *Journal of Economic Literature* 63(3).

Additional

- Kőszegi, Botond, and Adam Szeidl (2013). “A Model of Focusing in Economic Choice.” *Quarterly Journal of Economics* 128(1): 53–104.

4.5 Measuring Attention

Mandatory

- Caplin, Andrew, Dániel Csaba, John Leahy, and Oded Nov (2020). “Rational Inattention, Competitive Supply, and Psychometrics.” *Quarterly Journal of Economics* 135(3): 1681–1724.

- Bronchetti, Erin Todd, Judd B. Kessler, Ellen B. Magenheimer, Dmitry Taubinsky, and Eric Zwick (2023). “Is Attention Produced Optimally? Theory and Evidence from Experiments with Bandwidth Enhancements.” *Econometrica* 91(2): 669–707.

Additional

- Smith, Stephanie, Ian Krajbich, and Ryan Webb (2019). “Estimating the Dynamic Role of Attention via Random Utility.” *Journal of the Economic Science Association* 5: 97–111.
- Krajbich, Ian, Carrie Armel, and Antonio Rangel (2010). “Visual Fixations and the Computation and Comparison of Value in Simple Choice.” *Nature Neuroscience* 13(10): 1292–1298.
- Li, Shengwu, and Colin F. Camerer (2022). “Predictable Effects of Visual Saliency in Experimental Decisions and Games.” *Quarterly Journal of Economics* 137(3): 1849–1900.

5 Computational and Neuroeconomic Foundations

5.1 Psychophysics and Signal Detection

Classic

- Green, David M., and John A. Swets (1966). *Signal Detection Theory and Psychophysics*. New York: Wiley.
- Newsome, William T., Kenneth H. Britten, and J. Anthony Movshon (1989). “Neuronal Correlates of a Perceptual Decision.” *Nature* 341: 52–54.

Mandatory

- Glimcher, Paul W. (2011). *Foundations of Neuroeconomic Analysis*. New York: Oxford University Press. Chapter 4.
- Woodford, Michael (2020). “Modeling Imprecision in Perception, Valuation and Choice.” *Annual Review of Economics* 12: 579–601.

Additional

- Heeger, David J. (1997). “Signal Detection Theory.” Lecture notes, New York University.
- Girshick, Ahna R., Michael S. Landy, and Eero P. Simoncelli (2011). “Cardinal Rules: Visual Orientation Perception Reflects Knowledge of Environmental Statistics.” *Nature Neuroscience* 14: 926–932.
- Petzschner, Frederike H., Stefan Glasauer, and Klaas E. Stephan (2015). “A Bayesian Perspective on Magnitude Estimation.” *Trends in Cognitive Sciences* 19(5): 285–293.

5.2 Sequential Sampling and Evidence Accumulation

Classic

- Ratcliff, Roger (1978). “A Theory of Memory Retrieval.” *Psychological Review* 85(2): 59–108.
- Krajbich, Ian, Carrie Armel, and Antonio Rangel (2010). “Visual Fixations and the Computation and Comparison of Value in Simple Choice.” *Nature Neuroscience* 13(10): 1292–1298.

Mandatory

- Shadlen, Michael N., Timothy D. Hanks, Anne K. Churchland, Roozbeh Kiani, and Tianming J. Yang (2007). “The Speed and Accuracy of a Simple Perceptual Decision: A Mathematical Primer.” In Kenji Doya et al., eds., *Bayesian Brain: Probabilistic Approaches to Neural Coding*. Cambridge, MA: MIT Press.
- Ratcliff, Roger, and Gail McKoon (2008). “The Diffusion Decision Model: Theory and Data for Two-Choice Decision Tasks.” *Neural Computation* 20(4): 873–922.
- Fudenberg, Drew, Philipp Strack, and Tomasz Strzalecki (2018). “Speed, Accuracy, and the Optimal Timing of Choices.” *American Economic Review* 108(12): 3651–3684.

Additional

- Shadlen, Michael N., and Daphna Shohamy (2016). “Decision Making and Sequential Sampling from Memory.” *Neuron* 90(5): 927–939.
- Clithero, John A. (2018). “Improving Out-of-Sample Predictions Using Response Times and a Model of the Decision Process.” *Journal of Economic Behavior & Organization* 148: 344–375.
- Tajima, Satoshi, Jan Drugowitsch, Nisheet Patel, and Alexandre Pouget (2019). “Optimal Policy for Multi-Alternative Decisions.” *Nature Neuroscience* 22: 1503–1511.

5.3 Efficient Coding

Classic

- Laughlin, Simon B. (1981). “A Simple Coding Procedure Enhances a Neuron’s Information Capacity.” *Zeitschrift für Naturforschung C* 36: 910–912.

Mandatory

- Wei, Xue-Xin, and Alan A. Stocker (2015). “A Bayesian Observer Model Constrained by Efficient Coding Can Explain ‘Anti-Bayesian’ Percepts.” *Nature Neuroscience* 18: 1509–1517.
- Heng, Joseph A., Michael Woodford, and Rafael Polanía (2020). “Efficient Sampling and Noisy Decisions.” *eLife* 9: e54962.
- Frydman, Cary, and Lawrence J. Jin (2022). “Efficient Coding and Risky Choice.” *Quarterly Journal of Economics* 137(1): 161–213.

Additional

- Steiner, Jakub, and Colin Stewart (2016). “Perceiving Prospects Properly.” *American Economic Review* 106(6): 1601–1631.
- Juechems, Keno, Jan Balaguer, Bernhard Spitzer, and Christopher Summerfield (2021). “Optimal Utility and Probability Functions for Agents with Finite Computational Precision.” *Proceedings of the National Academy of Sciences* 118: e2002232118.

5.4 Neural Representation of Value and Choice

Classic

- Padoa-Schioppa, Camillo, and John A. Assad (2006). “Neurons in the Orbitofrontal Cortex Encode Economic Value.” *Nature* 441: 223–226.
- Rangel, Antonio, Colin Camerer, and P. Read Montague (2008). “A Framework for Studying the Neurobiology of Value-Based Decision Making.” *Nature Reviews Neuroscience* 9(7): 545–556.

Mandatory

- Camerer, Colin, George Loewenstein, and Drazen Prelec (2005). “Neuroeconomics: How Neuroscience Can Inform Economics.” *Journal of Economic Literature* 43(1): 9–64.
- Fehr, Ernst, and Antonio Rangel (2011). “Neuroeconomic Foundations of Economic Choice—Recent Advances.” *Journal of Economic Perspectives* 25(4): 3–30.

Additional

- Bernheim, B. Douglas (2009). “On the Potential of Neuroeconomics: A Critical (but Hopeful) Appraisal.” *American Economic Journal: Microeconomics* 1(2): 1–41.
- Glimcher, Paul W., and Ernst Fehr, eds. (2013). *Neuroeconomics: Decision Making and the Brain*, 2nd ed. Amsterdam: Academic Press.

6 Reference and Context Dependence

6.1 Reference Dependence

Classic

- Kahneman, Daniel, and Amos Tversky (1979). “Prospect Theory: An Analysis of Decision under Risk.” *Econometrica* 47(2): 263–292.
- Kahneman, Daniel, Jack L. Knetsch, and Richard H. Thaler (1990). “Experimental Tests of the Endowment Effect and the Coase Theorem.” *Journal of Political Economy* 98(6): 1325–1348.

- Tversky, Amos, and Daniel Kahneman (1991). “Loss Aversion in Riskless Choice: A Reference-Dependent Model.” *Quarterly Journal of Economics* 106(4): 1039–1061.
- Barberis, Nicholas C. (2013). “Thirty Years of Prospect Theory in Economics: A Review and Assessment.” *Journal of Economic Perspectives* 27(1): 173–196.

Mandatory

- Kőszegi, Botond, and Matthew Rabin (2006). “A Model of Reference-Dependent Preferences.” *Quarterly Journal of Economics* 121(4): 1133–1165.
- Dean, Mark, Benjamin Enke, Thomas Graeber, and Pietro Ortoleva (2026). “Reference Points as Information.” Working paper.

Additional

- Sugden, Robert (2003). “Reference-Dependent Subjective Expected Utility.” *Journal of Economic Theory* 111(2): 172–191.
- Kőszegi, Botond, and Matthew Rabin (2007). “Reference-Dependent Risk Attitudes.” *American Economic Review* 97(4): 1047–1073.

6.2 Salience, Focusing, and Relative Thinking

Classic

- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer (2012). “Salience Theory of Choice under Risk.” *Quarterly Journal of Economics* 127(3): 1243–1285.
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer (2013). “Salience and Consumer Choice.” *Journal of Political Economy* 121(5): 803–843.
- Kőszegi, Botond, and Adam Szeidl (2013). “A Model of Focusing in Economic Choice.” *Quarterly Journal of Economics* 128(1): 53–104.

Mandatory

- Bushong, Benjamin, Matthew Rabin, and Joshua Schwartzstein (2021). “A Model of Relative Thinking.” *Review of Economic Studies* 88(1): 162–191.
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer (2022). “Salience.” *Annual Review of Economics* 14: 521–544.

Additional

- Li, Shengwu, and Colin F. Camerer (2022). “Predictable Effects of Visual Salience in Experimental Decisions and Games.” *Quarterly Journal of Economics* 137(3): 1849–1900.

6.3 Normalization and Contextual Coding

Classic

- Soltani, Alireza, Benedetto De Martino, and Colin F. Camerer (2012). “A Range-Normalization Model of Context-Dependent Choice: A New Model and Evidence.” *PLoS Computational Biology* 8(7): e1002607.
- Louie, Kenway, Mel Win Khaw, and Paul W. Glimcher (2013). “Normalization Is a General Neural Mechanism for Context-Dependent Decision Making.” *Proceedings of the National Academy of Sciences* 110(15): 6139–6144.

Additional

- Gluth, Sebastian, Nadja Kern, Maël Lebreton, and Christian L. Vitali (2020). “Value-Based Attention but Not Divisive Normalization Influences Decisions with Multiple Alternatives.” *Nature Human Behaviour* 4: 634–645.
- Landry, Peter, and Ryan Webb. “Pairwise Normalization: A Neuroeconomic Theory of Multi-Attribute Choice.” Working paper.

6.4 Multi-Alternative Context Effects

Classic

- Huber, Joel, John W. Payne, and Christopher Puto (1982). “Adding Asymmetrically Dominated Alternatives: Violations of Regularity and the Similarity Hypothesis.” *Journal of Consumer Research* 9(1): 90–98.
- Simonson, Itamar (1989). “Choice Based on Reasons: The Case of Attraction and Compromise Effects.” *Journal of Consumer Research* 16(2): 158–174.
- Tversky, Amos, and Itamar Simonson (1993). “Context-Dependent Preferences.” *Management Science* 39(10): 1179–1189.

Additional

- Dumbalska, Tanya, Vickie Li, Konstantinos Tsetsos, and Christopher Summerfield (2020). “A Map of Decoy Influence in Human Multi-Alternative Choice.” *Proceedings of the National Academy of Sciences* 117(40): 25169–25178.
- Howes, Andrew, Paul A. Warren, Andrew J. Farmer, Wael El-Deredy, and Jason R. Lewis (2016). “Why Contextual Preference Reversals Maximize Expected Value.” *Psychological Review* 123(4): 368–391.
- Tsetsos, Konstantinos, Nick Chater, and Marius Usher (2016). “Economic Irrationality Is Optimal during Noisy Decision Making.” *Proceedings of the National Academy of Sciences* 113(11): 3102–3107.
- Natenzon, Paulo (2019). “Random Choice and Learning.” *Journal of Political Economy* 127(1): 419–457.

7 Simplicity and Complexity

7.1 Simplification and Sparse Representation

Classic

- Gabaix, Xavier (2014). “A Sparsity-Based Model of Bounded Rationality.” *Quarterly Journal of Economics* 129(4): 1661–1710.

Mandatory

- Gabaix, Xavier (2019). “Behavioral Inattention.” In *Handbook of Behavioral Economics*, Vol. 2, 261–343. Amsterdam: Elsevier.
- Kőszegi, Botond, and Filip Matějka (2020). “Choice Simplification: A Theory of Mental Budgeting and Naive Diversification.” *Quarterly Journal of Economics* 135(2): 1153–1207.

7.2 Decision Complexity: Sources and Measurement

Mandatory

- Kendall, Chad, and Ryan Oprea (2024). “On the Complexity of Forming Mental Models.” *Quantitative Economics* 15(1): 175–211.
- Enke, Benjamin, and Cassidy Shubatt (2023). “Quantifying Lottery Choice Complexity.” NBER Working Paper No. 31677.
- Gabaix, Xavier, and Thomas Graeber (2024). “The Complexity of Economic Decisions.” NBER Working Paper No. 33109.

Additional

- Shubatt, Cassidy, and Jeffrey Yang. “Tradeoffs and Comparison Complexity.” Working paper.
- Enke, Benjamin, and Thomas Graeber (2023). “Cognitive Uncertainty.” *Quarterly Journal of Economics* 138(4): 2021–2067.
- de Lara, Lucas. “A Closer Look at the Gambler’s Fallacy”. Working Paper.

7.3 Behavioral Consequences of Complexity

Classic

- Sonsino, Doron, Uri Benzion, and Galit Mador (2002). “The Complexity Effects on Choice with Uncertainty—Experimental Evidence.” *Economic Journal* 112(482): 936–965.

Mandatory

- Oprea, Ryan (2024). “Decisions under Risk Are Decisions under Complexity.” *American Economic Review* 114(12): 3789–3811.
- Enke, Benjamin, Thomas Graeber, Ryan Oprea, and Jeffrey Yang. “Behavioral Attenuation.” Forthcoming, *Journal of Political Economy*.
- Enke, Benjamin, Thomas Graeber, and Ryan Oprea (2025). “Complexity and Time.” *Journal of the European Economic Association* 23(5): 1838–1867.

Additional

- Gabaix, Xavier (2025). “A Theory of Complexity Aversion.” Working paper.
- Shubatt, Cassidy, and Jeffrey Yang. “Tradeoffs and Comparison Complexity.” Working paper.